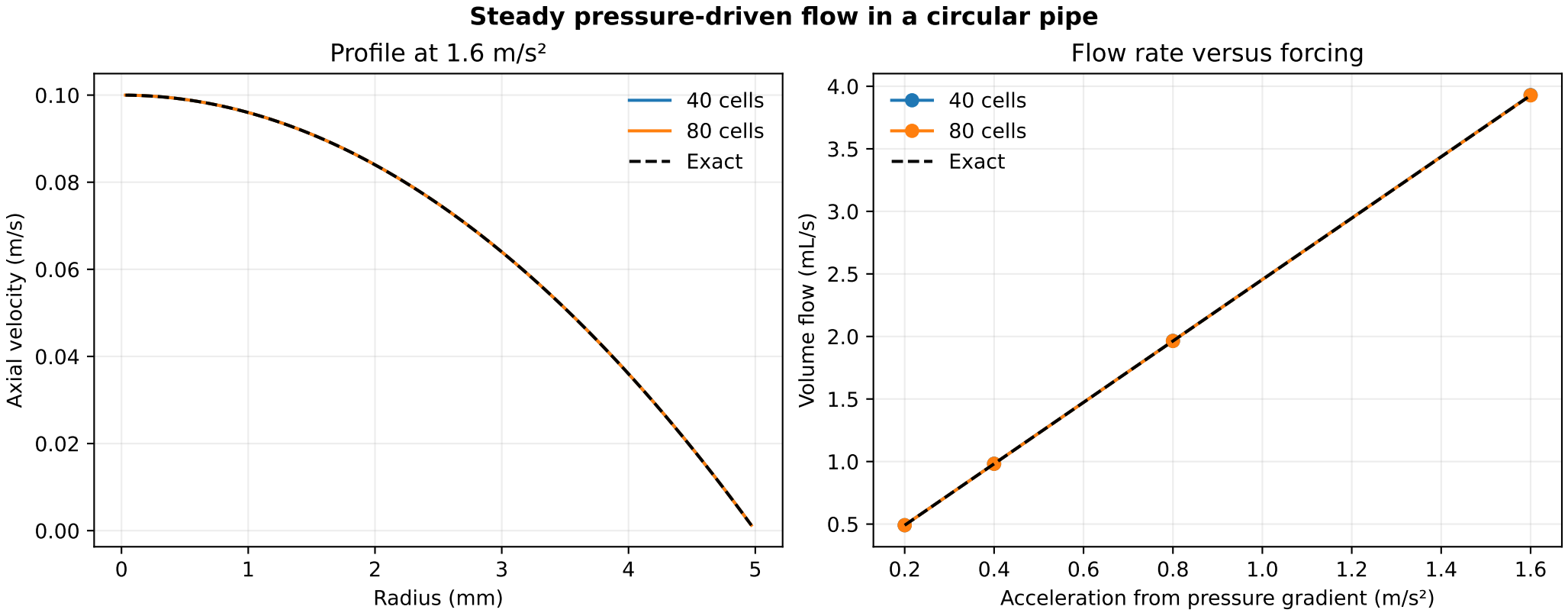
Steady pressure-driven flow in a circular pipe
Profile at 1.6 m/s²
Axial velocity (m/s)
Radius (mm)
40 cells
80 cells
Exact
0.00
0.02
0.04
0.06
0.08
0.10
0
1
2
3
4
5
Flow rate versus forcing
Volume flow (mL/s)
Acceleration from pressure gradient (m/s²)
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6